Changjiang Han

M.Sc. Student in Machine Learning, MBZUAI
Researcher, Cyber-Physical Intelligence Lab (CPIL)

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Research Interests

My research builds LLM agents that are capable, personalized, and learn reliably from feedback.

Agentic Reinforcement Learning: how agentic RL data scales across tasks, environments and trajectories.
Agent Memory: turning retrieved knowledge into parametric memory that needs no online retrieval.
LLM Personalization: the internal mechanisms that drive personalization in LLMs, and how to steer them.
Generative Recommendation: semantic-ID recommenders, from collaborative signals to deployment.
Multi-Agent Systems: what agents communicate, and how to audit and evaluate it.

Education

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) — Abu Dhabi, UAE
M.Sc., Machine Learning Department — Sept. 2025 – Present
GPA: 4.00/4.00. Supervisor: Prof. Xue (Steve) Liu.

Huazhong University of Science and Technology (HUST) — Wuhan, China
B.Eng. in Electronic and Information Engineering (Elite Class) — Sept. 2021 – June 2025
GPA: 3.88/4.00 (5/29).

Publications & Patents

* equal contribution.

First and Co-First Author

Changjiang Han, Bowei He, Ye Yuan, Weixu Zhang, Haolun Wu, Xue Liu, Jikun Kang. *From Lookups to Lessons: Turning Retrieved Knowledge into Parametric Memory*. (**ICLR 2027**, under review)

Changjiang Han, Qingyang Li, Yaqiang Zang, Jikun Kang, Pinghua Gong, Xue Liu, Bowei He. *Restoring Collaborative Signals in Semantic-ID Generative Recommendation via Personalized Natural Language*. (**WWW 2027**, under review)

Guo Tang*, Hanye Wu*, **Changjiang Han***, Qingyang Li, Ming Zhang, Xiangyu Qian, Yanchen Qiao, Huanjie Wang, Zhi Ma, Zhen Li, Yaqiang Zang, Pinghua Gong. *OxygenREC-v2: Internalizing Discrimination into Generative Recommendation*. (**KDD 2027**, under review)

Changjiang Han, et al. *An Automated Chemical Material Preparation Platform with Dual Collaborative Robotic Arms*. Chinese Invention Patent Application No. 202411034150.4, 2024.

Co-Authored

Bowei He, Yankai Chen, Xiaokun Zhang, **Changjiang Han**, Ye Yuan, Chong Li, Xue Liu. *How Data Scales in Agentic Reinforcement Learning: Laws and Synthesis Strategies*. In **NeurIPS 2026**.

Ye Yuan, Rui Song, Weien Li, Zeyu Li, Haochen Liu, Xiangyu Kong, **Changjiang Han**, Yonghan Yang, Zichen Zhao, Zixuan Dong, Fuyuan Lyu, Bowei He, Haolun Wu, Jikun Kang, Xue Liu. *QUACK: Questioning, Understanding, and Auditing Communicated Knowledge in Multimodal Social Deduction Agents*. In **EMNLP 2026**.

Weixu Zhang, Ye Yuan, **Changjiang Han**, Yuxing Tian, Zipeng Sun, Linfeng Du, Jikun Kang, Hong Kang, Xue Liu, Haolun Wu. *Preference Heads in Large Language Models: A Mechanistic Framework for Interpretable*

Personalization. In **ACL 2026**.

Ye Yuan, Weien Li, Rui Song, Zeyu Li, Haochen Liu, Xiangyu Kong, Zixuan Dong, Linfeng Du, Zipeng Sun, Weixu Zhang, Jiaxin Huang, **Changjiang Han**, Yonghan Yang, Zichen Zhao, Xiuyuan Hu, Haolun Wu, Yankai Chen, Fengran Mo, Jikun Kang, Bowei He, Dawn Song, Philip S. Yu, Xue Liu. *Discrete Diffusion Models: A Unified Framework from Tokenization to Generation*. arXiv preprint, 2026.

Dun Yuan, Fuyuan Lyu, Ye Yuan, Weixu Zhang, Bowei He, Jiayi Geng, Linfeng Du, Zipeng Sun, Yankai Chen, **Changjiang Han**, Jikun Kang, Xi Chen, Haolun Wu, Xue Liu. *Beyond Message Passing: A Semantic View of Agent Communication Protocols*. arXiv preprint, 2026.

Experience

AMD Remote
Research Intern Nov. 2026 – Present
Open Source & AI, AMD. Mentor: Xunzhuo Liu. Working on vLLM Semantic Router, the vLLM project's system that routes each LLM request to the most suitable model, an open-source project with 6.4K GitHub stars and adopted by Microsoft, Red Hat and Dell. Also working on the application and development of its Decision models, a family of open foundation decision models trained and served on AMD Instinct GPUs.

JD.com Beijing, China
Research Intern, Tech Genius Team (TGT) Apr. 2026 – Sept. 2026
Smart Marketing and Recommendation Department, JD Retail. Mentor: Qingyang Li. Worked on the OxygenREC generative recommendation architecture, including reasoning instructions for end-to-end generative recommendation. Co-first author of OxygenREC-v2; first author of SID-NL.

State Key Laboratory of Brain-Machine Intelligence, Zhejiang University Hangzhou, China
Research Intern May 2024 – Sept. 2024
Supervisor: Prof. Gang Pan. Worked on EEG data preprocessing, neuron parameter design for spiking neural networks, and hybrid SNN–ANN architectures.

Automated Chemistry Laboratory, Dian Group, HUST Wuhan, China
Project Leader June 2023 – June 2025
Led a two-year project building an AI-scheduled, vision-guided dual-arm robotic platform for automated chemical preparation, from design to deployment; designed the robot control.

Awards

Outstanding Undergraduate Thesis Award, HUST (Top 3%) 2025
Outstanding Graduate Award, HUST (Top 15%) 2025
First Prize (Hubei Province), China Undergraduate Mathematical Contest in Modeling (Top 10%) 2023
Honorable Mention, Mathematical Contest in Modeling (MCM) (Top 15%) 2022

Skills

Programming: Python, C++, Assembly; PyTorch, ROS.
Parallel Computing: multi-node distributed training on GPU clusters (32× A100 / B200).
Hardware & Communications: embedded circuit design; wireless communication.
Languages: Mandarin Chinese (native); English (fluent).

References

Prof. Xue (Steve) Liu (M.Sc. Supervisor)
Professor, MBZUAI and McGill University
FCAE, FIEEE

Qingyang Li (Internship Mentor)
Senior Research Scientist, JD Retail, JD.com